H-B-451G
February 16, 1995
Superseding
H-B-451F
September 5, 1991

FEDERAL SPECIFICATION

BRUSH, RADIATOR BRONZING

The General Services Administration has authorized the use of this federal specification for all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers two types of radiator bronzing brushes.

1.2 Classification.

1.2.1 Types. Radiator bronzing brushes shall be of the following types as specified (see 6.2):

Type I - Short handle
Type II - Long handle

1.2.2 Size. Radiator bronzing brushes shall be of the following sizes as specified (see 6.2):

25 mm, (1 in.)
38 mm. (1-1/2 in.)
51 mm. (2 in.)

2. APPLICABLE DOCUMENTS

2.1 The following document of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specification:

PPP-B-1880 - Brushes; Preparation for Delivery of

(Copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available from the following address: GSA Specifications Section (3FBP-W), Suite 8100, 470 L'enfant Plaza SW, Washington, DC 20407.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: General Services Administration, GSA Center (10FTE), Auburn, WA 98001.

DISTRIBUTION STATEMENT: Approved for public release; Distribution is unlimited.

FSC 8020

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Military Standards:

MIL-STD 105 – Sampling Procedures and Tables for Inspection by Attributes

(Copies of Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from the contracting officer.)

Federal Regulations:

29 CFR 110.1200 – Hazard Communications Standard

(Copies of Specifications and Standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 2200 Mill Rd., Alexandria, VA 22314.)

American Society for Testing and Materials:

ASTM D13 – Specification for Spirits of Turpentine

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials and Components. Materials and components shall conform to the requirements specified herein.

3.1.1 Hair and Bristle. The brushes shall be either 100 percent soft hog bristle, unbleached and undyed, or a blend of 66-2/3 percent goat hair and 33-1/3 percent soft hog bristle, unbleached and undyed. The bristle shall pass the straightness test specified in 4.4.4. A certificate of compliance from the supplier that the bristle meets all requirements (stiffness and sterilization) may be accepted. The contractor shall provide certification that color(s) has not been added to the bristles. No more than 5 bristles shall be removed when pull tested, per 4.4.2.2(c),

3.1.2 Handle. The handle shall be made of close-grained hard-wood or a synthetic material with all sprues and runners removed. The synthetic handle shall not crack or break when tested as in 4.4.3, and shall show no perceptible change when tested as in 4.4.2.

3.1.3 Ferrule. The ferrule shall be made from corrosion-resistant metal or metal made corrosion-resistant by electro-plating.

3.1.4 Setting compound. The setting compound shall be a material which shall withstand the test specified in 4.4.2 without softening, deteriorating, or loosening of bristle in the compound.

3.1.5 Filler strips. Filler strips shall be made of wood, metal, or plastic and shall not soften or otherwise change when tested as in 4.4.2. Cardboard filler strips are allowed for machine made brushes.

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3.2 Construction

3.2.1 Physical requirements. The brushes shall conform to the physical requirements in Table I.

TABLE I. Physical requirements for bronzing brushes

Characteristic	Size		
	25 mm. (1 in.)	38 mm. (1-1/2 in.)	51 mm. (2 in.)
Brush part:			
Width inside ferrule, min., mm.(in.)	25 (31/32)	37 mm (1-15/32)	50 mm (1-31/32)
Thickness inside ferrule, min., mm. (in.)	7 (9/32)	7 (9/32)	9 (11/32)
Hair and Bristle:			
Weight in finished brush, min., gm. (oz.)	4 (5/32)	8 (17/64)	10 (13/32)
Length outside of ferrule, min., mm. (in.)	32 (1-1/4)	35 (1-3/8)	38 (1-1/2)
Overall length of brush:			
Type I, min., cm. (in.)	23.8 (9-3/8)	25.4 (10)	26.4 (10-3/8)
Type II, min. cm. (in.)	25.4 (10)	31.4 (12-3/8)	32.7 (12-7/8)

3.2.2 Ferrule. The ferrule shall be made of the materials specified in 3.1.3 and shall have an end lap of 3 mm. (1/8 in.) and shall be interlocked and soldered, lock-seamed, or welded. The ferrule metal shall have a minimum wall thickness of .5 mm. (0.010 in.)

3.2.3 Set. The butt ends of the bristles and/or hair filaments shall be set in a setting compound conforming to 3.1.4. When tested as in 4.4.2.2, there shall be no more than a total of five (5) bristles and/or hair filaments pulled loose during the tensile test.

3.2.4 Assembly. The ferrule shall be secured to the handle by crimping or nailing.

a) 25 mm. (1 in.) and 38 mm. (1-1/2 in.) brushes shall have at least one nail on each side of the brush.

b) 51 mm. (2 in.) brushes shall have a minimum of two nails on each side. The nails shall be equidistant from the sides of the ferrule and shall be separated by approximately one-half the brush width.

c) Nails shall be corrosion resistant and shall be a minimum of 10 mm. (3/8 in.) in length.

d) Brushes with ferrules secured to the handle by crimping shall not become loosened when tested as in 4.4.3.

3.2.5 Trim. The brushes shall be formed to a square top. They shall be trimmed evenly and uniformly to the finished length outside of the ferrule.

3.2.6 Marking for identification. Unless otherwise specified (see 6.2) in addition to any special marking required by the contract or order, each brush shall be marked with the contractor's name or trademark and with the National Stock Number.

3.2.7 Workmanship. The finished brushes shall be free of residual debris and have no sharp edges, splinters, burrs, or any other defects which may affect appearance or serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein using facilities acceptable to the Government. The Government reserves the right to perform any inspection herein when deemed necessary to assure that the brushes conform to prescribed requirements.

4.2 Inspection of preparation for delivery. An examination shall be made to determine compliance with 3.2 and section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

4.3 Inspection of the end item. Lots shall be assembled and randomly sampled in accordance with MIL-STD 105, inspection level I, with an acceptable quality level (AQL) of 2.5 percent defective. Sample brushes shall be examined for the following attributes:

Bristles and hair
Ferrules
Handles
Construction and assembly
Set
Trim
Workmanship
Identification markings

Any brush in the sample containing one or more visual or dimensional defects shall be considered a defective brush.

4.3.1 Sampling for testing of the end item. A random sample of brushes shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 2.5 percent defective.

4.4 Testing procedures. Samples, selected in accordance with 4.3.1, shall be subjected to the following tests. Failure to pass any test shall be cause for rejection of the lot. All tests shall be evaluated for conformance to the requirements in section 3.

4.4.1 Weight of bristle and/or hair. After removal of the ferrule, the setting compound shall be removed from the bristle and/or hair by any practical means which does not injure the bristle/hair. The weight of bristles/hair shall be determined for compliance with physical requirements.

4.4.2 Solvent resistance. Solvent resistance tests shall be performed at room temperature.

4.4.2.1 Handle (Organic coating). Brush handles with organic coating shall be tested as follows:

A. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in commercial mineral spirits. After 30 minutes, the handle shall be removed from the mineral spirits and examined. There shall be no wrinkling, blistering, or separation of the coating.

B. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in turpentine conforming to ASTM D13. After 30 minutes, the handle shall be removed from the turpentine and examined. There shall be no wrinkling, blistering, or separation of the coating.

C. The brush handle shall be dried for 2 hours and the organic coating shall be examined. The coating shall not be wrinkled, blistered, or separated from the handle and the coating shall not be soft.

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4.4.2.2 Base material, filler strips, synthetic handles. The brush shall be tested for solvent resistance using commercial lacquer thinner. (NOTE: The organic coating on the handle may be removed by this solvent. The objective is to test solvent resistance of the handle base material, filler strips, and the setting compound.)

The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in lacquer thinner for 1 hour. Synthetic handles shall be immersed to a depth of one-half of the handle length. The brush shall be removed and examined as follows:

A. Handle and ferrule - As soon as the brush is removed from the lacquer thinner, the brush shall be dried for 2 hours and examined for deterioration of the handle base material.

B. Filler strips - When the brush is removed from the lacquer thinner, filler strips shall be examined for softening, swelling, or other deleterious change.

C. Setting compound. The setting compound shall be examined for softening, blistering, cracking, crazing or other deleterious effects by the solvent. The brush shall be suspended by the handle in a holding device, and a group of bristles 6 mm. (1/4 in.) in diameter shall be given a tensile pull of 4.5 k. (10 lb.) away from the ferrule. Repeat this tensile pull test at random places one time for each 2.5 cm. (1 inch) of brush width. During the tensile pull, no more than five (5) bristles/filaments shall be removed. After thorough drying the brush shall be examined for solvent damage to the handle base material (synthetic handles). The ferrule shall be removed and the setting compound shall be examined for deterioration, loosening of the bristles, or other solvent effects.

4.4.3 Drop test. Synthetic brushes shall be dropped twice from a height of three meters (10 feet) onto a concrete surface. Brushes, in which the handles are fastened by means other than nailing, shall be similarly dropped as assembled and examined for compliance with 3.1.3 and 3.2.1.

4.4.4 Straightness. The assembled brush head shall be placed so that the bristle is immersed in 60 degree C (140 degree F) water for five (5) minutes. Excess water shall be shaken out. The brush shall be hanged with bristle down to dry completely. The flare shall be measured by aligning the ferrule portion of the brush as shown in figure 1. Flare measurement is limited to the outline of the main body of the brush. Stray bristles shall be disregarded.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging and packing shall be A, B, C, or otherwise specified in the contract or order (see 6.2). The marking shall be civil, military, or as specified (see 6.2).

6. Notes

6.1 Intended use. The brushes covered by this specification are intended for use in painting items having restricted hard-to-reach areas.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement of documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1).
(c) Levels of packaging and packing required (see 5.1).
(d) Marking required (see 5.1).

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PREPARING ACTIVITY:

GSA - FSS

CIVIL AGENCIES COORDINATING:

GSA - FSS

JUS - FPI

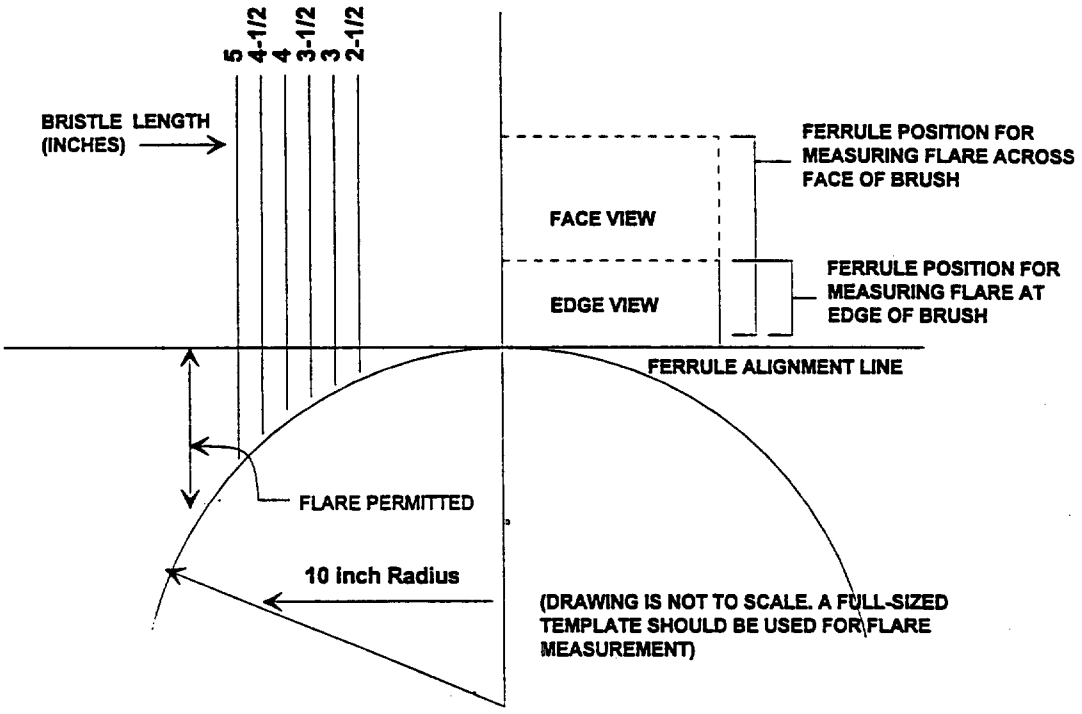


Figure 1. Flare measurement

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